Block practice · Encryption, network security, ethics and new technology

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

What encryption is and why we use it

Keeping a network secure: access rights, physical security and authentication

Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Ethics of technology: privacy, inclusion and professionalism

Emerging tech: quantum, DNA computing, AI and nanotechnology

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Practice workbook · What encryption is and why we use it

What encryption is and why data is encrypted.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-17

1. Question 3 · part b · 1 mark

2020 NOV · Paper 02

Worked-example retry · Attempt part b only.

3 Several encryption algorithms have been developed.

(a) State what is meant by the term 'encryption'.

(1)

.....

.....

(b) Give **one** reason why data may need to be encrypted.

(1)

.....

.....

(c) (i) Perform a Rail Fence cipher encryption of the text **COMPUTATIONAL** using a key of 4.

Show your working.

(4)

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Encrypted text:

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6

P 6 1 8 8 5 A 0 6 1 6

1. Question 3 · part b · 1 mark · Working and answer

Working and answer

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2. Question 6 · part a.iii · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.iii only.

6 A train company uses ticket vending machines at each station.

(a) The machines use embedded systems.

(i) Explain **one** benefit of using an embedded system in these machines.

(2)

(ii) Customers use a touch screen to select their destination. They can pay by cash or bank card. Their tickets and a receipt are printed.

The touch screen is controlled by an embedded system.

Give **two** other hardware components in the ticket machine that are controlled by embedded systems.

(2)

1 .....

2 .....

(iii) The ticket machine uses data encryption when a customer pays using a bank card.

State why data encryption is used in this case.

(1)

.....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

20

P 6 1 8 8 0 A 0 2 0 2 4

2. Question 6 · part a.iii · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.iii only.

(b) Compare **four** features of high-level and low-level programming languages.

(4)

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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2. Question 6 · part a.iii · 1 mark · Working and answer

Working and answer

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3. Question 3 · part a · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part a only.

3 Several encryption algorithms have been developed.

(a) State what is meant by the term 'encryption'.

(1)

(b) Give **one** reason why data may need to be encrypted.

(1)

(c) (i) Perform a Rail Fence cipher encryption of the text **COMPUTATIONAL** using a key of 4.

Show your working.

(4)

Encrypted text:

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3. Question 3 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part a · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a only.

4 Several encryption algorithms have been developed.

(a) Identify what is meant by **encryption**.

(1)

- ☒ **A** Conversion of ciphertext into plaintext
- ☒ **B** Conversion of plaintext into ciphertext
- ☒ **C** Conversion of code into data
- ☒ **D** Conversion of data into information

(b) **Figure 1** shows a Pigpen cipher grid.

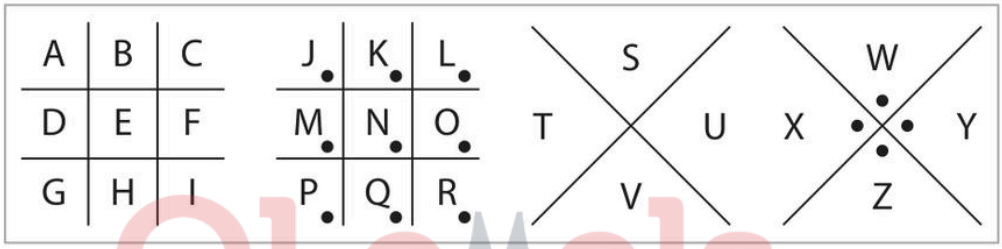


Figure 1

Complete the table by adding the symbol for each letter in the word **MAY**.

(3)

Letter	M	A	Y
Symbol			

(c) The message THE ENEMY IS NEAR is encoded using a different encryption algorithm.

Figure 2 shows the first stage of the encryption process.

T								I					
	H						Y		S				
		E				M				N			
			E		E						E		R
				N								A	

Figure 2

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

4. Question 4 · part a · 1 mark · Working and answer

Working and answer

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5. Question 1 · part b.i · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.i only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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2

P 7 2 9 3 7 A 0 2 2 4

5. Question 1 · part b.i · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

(2)

1 .....

.....

2 .....

.....

(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

(1)

.....

.....

(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

(1)

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

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5. Question 1 · part b.i · 1 mark · Working and answer

Working and answer

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Practice workbook · Keeping a network secure: access rights, physical security and authentication

Why networks need protecting and the measures used: access rights, password policies, physical protection of servers, authentication methods and the difference between validation and authentication.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-07

1. Question 4 · part c · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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1. Question 4 · part c · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(d) Marie is a researcher in nanotechnology.

(i) State what is meant by the term **nanotechnology**.

(1)

(ii) Give **one** application of nanotechnology.

(1)

(e) Marie has a mobile phone.

The address of the device is:

2024:0eb8:3c4d:0015:0000:0000:1a2f:1a2b

Describe the relationship between the structure of this address and the total number of bits required to store it.

(3)

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1. Question 4 · part c · 2 marks · Working and answer

Working and answer

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2. Question 2 · part b · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

2 Zafer and Robert work for a company that makes washing machines.

(a) Zafer writes user manuals for the washing machines.

He stores these documents in the cloud.

Zafer and the cloud storage provider share responsibility for data security.

State **one** area of responsibility for each of them.

(2)

Zafer

.....

.....

Cloud storage provider

.....

.....

(b) Robert programs the robots that make the parts for the washing machines.

Zafer writes the user manuals.

Robert and Zafer have different levels of access to folders and files on the company network.

The different levels of access are Read, Write, Execute and None.

Complete the table to show the levels of access that each has.

(2)

	Robert	**Zafer**
Washing machine design drawings		
A folder of manuals for new machines		

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 1 8 8 0 A 0 4 2 4

2. Question 2 · part b · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

(c) Zafer uses a browser to view pages on Pearson's website by typing a uniform resource locator into a browser. This is shown in **Figure 1**.



Figure 1

Pearson's machine	Zafer's machine
MAC address: 03-57-00-EC-4B-30	MAC address: 00-15-00-BC-9A-90
IPv4 address: 2.20.38.113	IPv4 address: 192.168.1.78
IPv4 Subnet Mask: 255.255.0.0	IPv4 Subnet Mask: 255.255.255.0
Website domain: www.pearson.com	Web browser
Main web page: https://www.pearson.com/uk/	Uniform resource locator: www.pearson.com

A domain name server is used in this process.

Identify the input to and the output from the domain name server.

(2)

Input

Output

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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P 6 1 8 8 0 A 0 5 2 4

5

Turn over ▶

2. Question 2 · part b · 2 marks · Working and answer

Working and answer

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3. Question 2 · part a · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a only.

2 Danielle manages network security for a small business.

(a) Identify **one** action Danielle could take to protect the **network** from unauthorised access.

(1)

- ☒ **A** Give each user a separate account
- ☒ **B** Enforce a strong password policy
- ☒ **C** Make a backup plan
- ☒ **D** Encrypt sensitive files

(b) Danielle has received this email.

She is a customer of SafePayment.com but thinks this is a phishing attempt.

From: SafePayment.accounts@5afepayment.com
To: Danielle616
Subject: Account restricted (case SP-0011312-2021-06)

Dear Customer,

We have noticed some unusual activity on your account, so have stopped all payments.

Please use this link to log in and check your account.

Log in

Your normal log in will stoop working until you have done this.

Regards
SafePayment fraud prevention team.

(i) Danielle has circled the words **Dear Customer**.

Give **one** reason why the use of these words might make Danielle suspect that the email is a phishing attempt.

(1)

(ii) Draw circles around **two other** aspects of the email that might make Danielle suspect it is a phishing attempt.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

6

P 6 9 2 9 7 A 0 6 2 0

3. Question 2 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 1 · part b.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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4. Question 1 · part b.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

(2)

1 .....

.....

2 .....

.....

(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

(1)

.....

.....

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(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

(1)

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

4. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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5. Question 4 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

4 Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

Type	Example	Number of bits
IPv4	192.169.0.3	
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	

(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

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DO NOT WRITE IN THIS AREA

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5. Question 4 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

(ii) Some people confuse Ethernet® and Wi-Fi.

Describe what is meant by the term **Ethernet®**.

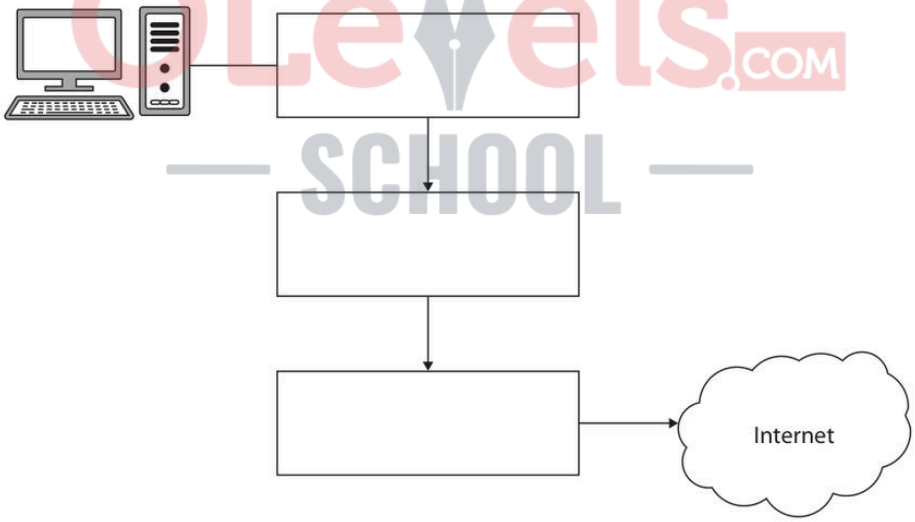
(2)

(d) A single physical box connects a desktop computer to the Internet.

The box incorporates three different components.

Complete the diagram to show the names of the components in the correct order.

(3)



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 2 9 3 7 A 0 1 2 2 4

5. Question 4 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

DO NOT WRITE IN THIS AREA

(e) Users enter passwords when logging onto a network and when creating accounts using a web page.

Describe **one** difference between validation and authentication.

(2)

(Total for Question 4 = 12 marks)

Working and answer

OLevels.com SCHOOL

6. Question 1 · part b · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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6. Question 1 · part b · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(c) **Figure 1** shows a network topology.

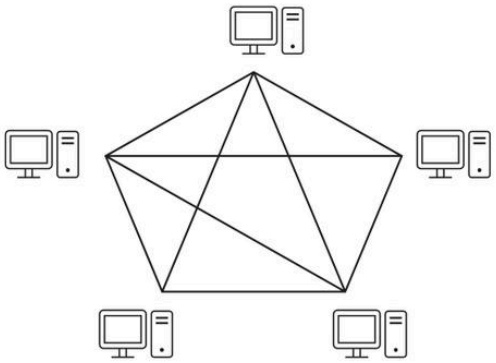


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

(2)

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6. Question 1 · part b · 1 mark · Working and answer

Working and answer

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7. Question 4 · part a.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a.ii only.

4 A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

(2)

1 .....

.....

2 .....

.....

(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

(2)

.....

.....

.....

.....

(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

(2)

1 .....

.....

2 .....

.....

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

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7. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Methods for finding weaknesses (penetration testing, ethical hacking, analysis tools, policy review) and protecting software (code reviews, audit trails, modular testing, secure design).

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-10

1. Question 1 · part e · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

1 Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

(4)

Statement	DNS	PAN	WAN	WWW
A network of networks **not** limited to a single location	☒	☒	☒	☒
A network that connects devices within the range of a single user	☒	☒	☒	☒
A set of hypertext documents accessed through the HTTP protocol	☒	☒	☒	☒
A system that maps domain names to IP addresses	☒	☒	☒	☒

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

(2)

1 .....

.....

2 .....

.....

(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

(1)

☒ **A** HTTP

☒ **B** IMAP

☒ **C** POP3

☒ **D** SMTP

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1. Question 1 · part e · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(d) A library sets up a network using a star topology.

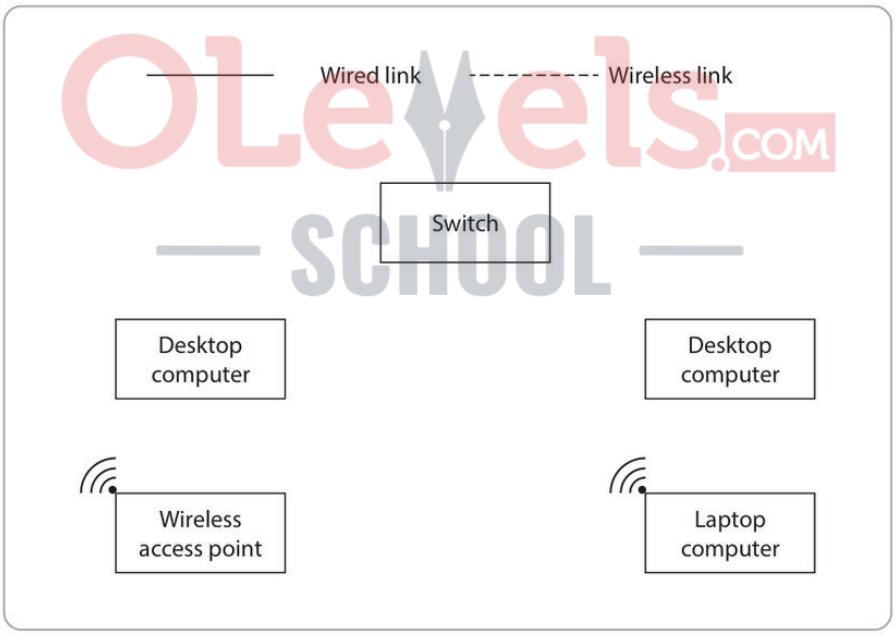
The library is connecting:

- two desktop computers with wired network interface cards
- a laptop with a wireless network interface card
- a switch
- a wireless access point.

Complete the diagram to show how the devices are connected.

Use a solid line to indicate a wired link and a dashed line to indicate a wireless link.

(3)



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1. Question 1 · part e · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

(e) IT teams carry out network penetration testing.

Describe the purpose of penetration testing.

(2)

(Total for Question 1 = 12 marks)

DO NOT WRITE IN THIS AREA

Working and answer

OLevels.com SCHOOL

2. Question 5 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5 Programmers work as part of a team to develop software.

(a) They use truth tables to determine if an expression evaluates to true or false.

Complete this truth table for the Boolean expression

P AND (Q OR R)

You may not need to use all rows.

(3)

P	Q	R	Q OR R	P AND (Q OR R)

OLevels.COM SCHOOL

2. Question 5 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

(b) The team is working on software for a ride-on lawn mower.

The lawn mower must not start unless it is safe for the driver.

- The brake (B) must be engaged.
- The driver's seat sensor (S) must be engaged.
- The blade safety lock (L) must be removed.

Construct a Boolean expression, using AND, OR, and NOT with the letters B, S and L to show the safe starting condition.

(2)

(c) The team uses code reviews as part of the software development process.

State **two** ways in which code reviews can help protect software systems from cyber attacks.

(2)

1 .....

2 .....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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14

P 6 1 8 8 4 R R A 0 1 4 2 0

2. Question 5 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

(d) The team needs to protect its intellectual property rights over the systems that it develops.

Give **three** types of legal protection the team could use.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(Total for Question 5 = 10 marks)

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Working and answer

3. Question 2 · part d.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part d.i only.

2 Danielle manages network security for a small business.

(a) Identify **one** action Danielle could take to protect the **network** from unauthorised access.

(1)

- ☒ **A** Give each user a separate account
- ☒ **B** Enforce a strong password policy
- ☒ **C** Make a backup plan
- ☒ **D** Encrypt sensitive files

(b) Danielle has received this email.

She is a customer of SafePayment.com but thinks this is a phishing attempt.

From: SafePayment.accounts@5afepayment.com
To: Danielle616
Subject: Account restricted (case SP-0011312-2021-06)

Dear Customer,

We have noticed some unusual activity on your account, so have stopped all payments.

Please use this link to log in and check your account.

Log in

Your normal log in will stoop working until you have done this.

Regards
SafePayment fraud prevention team.

(i) Danielle has circled the words **Dear Customer**.

Give **one** reason why the use of these words might make Danielle suspect that the email is a phishing attempt.

(1)

(ii) Draw circles around **two other** aspects of the email that might make Danielle suspect it is a phishing attempt.

(2)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6

P 6 9 2 9 7 A 0 6 2 0

3. Question 2 · part d.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part d.i only.

(c) Some employees use company laptops in public places.

Danielle is concerned that shoulder surfing could pose a risk to security.

(i) Describe what is meant by **shoulder surfing**.

(2)

(ii) Explain **one** way to prevent shoulder surfing.

(2)

(d) Danielle wants to protect the network from cyberattacks.

(i) Identify the purpose of penetration testing.

(1)

- ☒ **A** To decompile executable code
- ☒ **B** To identify security vulnerabilities
- ☒ **C** To test the strength of an encryption algorithm
- ☒ **D** To enforce a network user policy

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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3. Question 2 · part d.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part d.i only.

(ii) Danielle is sent a software patch for the network operating system but is too busy to install it straight away.

Explain why this delay could pose a threat to the security of the network.

(2)

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(Total for Question 2 = 11 marks)

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Working and answer

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4. Question 6 · part c.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.i only.

6 Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

Figure 3 shows a network topology.

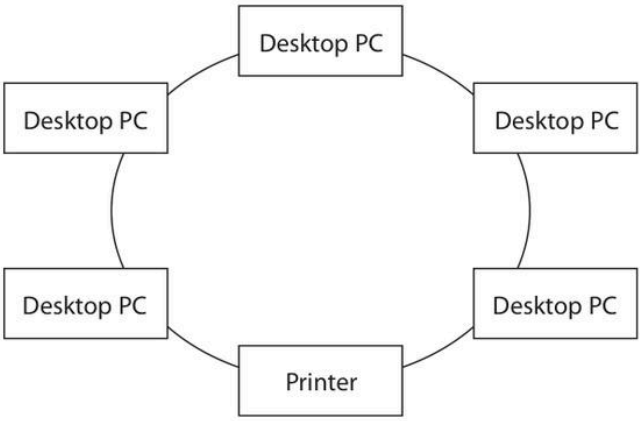


Figure 3

(i) Explain **one** benefit of this network topology. **(2)**

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet. **(2)**

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P 7 2 5 3 8 A 0 1 8 2 0

4. Question 6 · part c.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.i only.

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(b) Santiago works on his laptop whilst travelling by train.

There is a free Wi-Fi connection on the train, but Santiago doesn't use it.

He prefers to set up a network between his smartphone and his laptop to connect to the internet.

(i) Name this type of network.

(1)

(ii) Explain **one** advantage for Santiago of using the network he has set up to connect to the internet, rather than the free Wi-Fi connection.

(2)

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(c) Santiago uses audit trails to help protect the network.

(i) State what is meant by an **audit trail**.

(1)

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

(1)

4. Question 6 · part c.i · 1 mark · Working and answer

Working and answer

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5. Question 6 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

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Figure 3 shows a network topology.

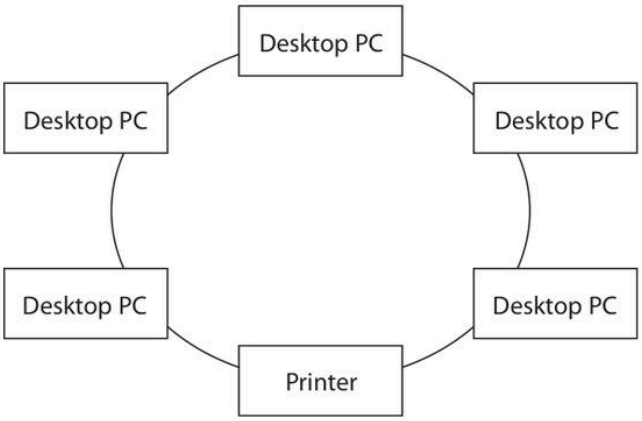


Figure 3

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5. Question 6 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

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(b) Santiago works on his laptop whilst travelling by train.

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(1)

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

(1)

5. Question 6 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

(d) Santiago wants to find and fix network vulnerabilities before the reputation of the company suffers.

Discuss the methods he can use.

You should consider:

- ethical hacking
- commercial analysis tools
- review of network and user policies.

(6)

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(Total for Question 6 = 15 marks)

5. Question 6 · part c.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 6 · part d · 6 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

6 Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

Figure 3 shows a network topology.

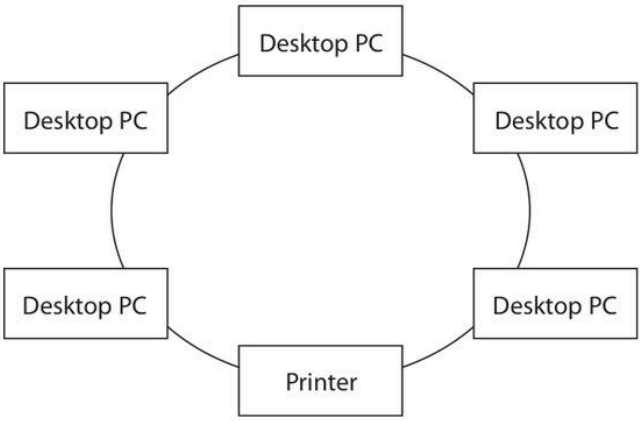


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P 7 2 5 3 8 A 0 1 8 2 0

6. Question 6 · part d · 6 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA

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(b) Santiago works on his laptop whilst travelling by train.

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He prefers to set up a network between his smartphone and his laptop to connect to the internet.

(i) Name this type of network.

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(2)

OLevels.COM SCHOOL

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(i) State what is meant by an **audit trail**.

(1)

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

(1)

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2022 JUN · Paper 01

Further practice · Attempt part d only.

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Discuss the methods he can use.

You should consider:

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(6)

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(Total for Question 6 = 15 marks)

6. Question 6 · part d · 6 marks · Working and answer

Working and answer

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7. Question 1 · part d · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part d only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

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7. Question 1 · part d · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part d only.

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(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

(2)

1 .....

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2 .....

.....

(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

(1)

.....

.....

(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

(1)

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

OLevels.com SCHOOL

7. Question 1 · part d · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part d only.

(d) Programmers write and test code.

They use modular testing.

State what is meant by the term **modular testing**.

(1)

.....

.....

(e) Computer scientists are viewed as professionals, in the same way that doctors are viewed as professionals.

Give **two** ways that computer scientists can demonstrate professionalism.

(2)

1 .....

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2 .....

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(Total for Question 1 = 10 marks)

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OLevels.COM SCHOOL

Working and answer

8. Question 4 · part a.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a.iii only.

4 A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

(2)

1 .....

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2 .....

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(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

(2)

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(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

(2)

1 .....

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2 .....

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8. Question 4 · part a.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a.iii only.

(b) The battery storage system stores readings as signed 8-bit binary integers in **sign and magnitude** format.

Complete the table to show the 8-bit binary pattern and denary value for the lowest and the highest values that can be stored.

(2)

	8-bit binary pattern	Denary value
Lowest value		
Highest value		

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P 7 8 6 3 7 A 0 1 1 2 0

8. Question 4 · part a.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Ethics of technology: privacy, inclusion and professionalism

Barriers to equal access, reluctance to use online services, privacy concerns of surveillance, and how computing professionals show ethical behaviour.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: CSC-T56-15

1. Question 4 · part f · 6 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part f only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

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1. Question 4 · part f · 6 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part f only.

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(d) Marie is a researcher in nanotechnology.

(i) State what is meant by the term **nanotechnology**.

(1)

(ii) Give **one** application of nanotechnology.

(1)

(e) Marie has a mobile phone.

The address of the device is:

2024:0eb8:3c4d:0015:0000:0000:1a2f:1a2b

Describe the relationship between the structure of this address and the total number of bits required to store it.

(3)

OLevels.com SCHOOL

1. Question 4 · part f · 6 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part f only.

(f) The research facility is considering introducing cameras and facial recognition software to monitor the work and movement of people on and around its site.

Discuss the privacy issues that need to be considered when deciding whether or not to install this new system.

(6)

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(Total for Question 4 = 16 marks)

1. Question 4 · part f · 6 marks · Working and answer

Working and answer

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2. Question 6 · part b · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

6 (a) Two emerging technologies are DNA computing and quantum computing.

(i) Explain **one** difference between DNA computing and traditional computing.

(2)

(ii) Give a definition of the term **superposition** in quantum computing.

(1)

(b) Ethically, everyone should be able to access the benefits of technology.

Give **two** reasons why this may not be possible.

(2)

1

2

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OLevels.COM SCHOOL

16

P 6 6 4 8 9 A 0 1 6 2 0

2. Question 6 · part b · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

(c) Widespread use of computer technology has a negative environmental impact.

Discuss the negative environmental impact of using computers and actions that could be taken to reduce it.

(6)

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(Total for Question 6 = 11 marks)

2. Question 6 · part b · 2 marks · Working and answer

Working and answer

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3. Question 5 · part a.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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3. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 1 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

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4. Question 1 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

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(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

(2)

1 .....

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2 .....

.....

(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

(1)

.....

.....

(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

(1)

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

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4. Question 1 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

(d) Programmers write and test code.

They use modular testing.

State what is meant by the term **modular testing**.

(1)

.....

.....

(e) Computer scientists are viewed as professionals, in the same way that doctors are viewed as professionals.

Give **two** ways that computer scientists can demonstrate professionalism.

(2)

1 .....

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2 .....

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(Total for Question 1 = 10 marks)

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Working and answer

5. Question 5 · part b.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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1 .....

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2 .....

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(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

5. Question 5 · part b.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

(2)

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

(2)

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

(1)

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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P 7 8 6 3 7 A 0 1 4 2 0

5. Question 5 · part b.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Emerging tech: quantum, DNA computing, AI and nanotechnology

Definitions, characteristics and applications of quantum computing, DNA computing, artificial intelligence and nanotechnology, including discussing AI's impact.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: CSC-T56-17

1. Question 2 · part d · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part d only.

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2 Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

(2)

1 .....

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2 .....

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(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

(1)

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

(2)

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OLevels.COM SCHOOL

1. Question 2 · part d · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part d only.

(c) The CPU performs the fetch-decode-execute cycle.

(i) Give **three** factors that affect the performance of the CPU.

(3)

1 .....

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2 .....

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3 .....

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(ii) Describe how the address bus is used during the fetch-decode-execute cycle.

(2)

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(d) Classical computers store data using bits that exist in one of two binary states, 1 or 0.

Quantum computers store data in a superposition of states.

Give **two other** characteristics of quantum computers.

(2)

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(Total for Question 2 = 12 marks)

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1. Question 2 · part d · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 6 · part c · 6 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

6 A train company uses ticket vending machines at each station.

(a) The machines use embedded systems.

(i) Explain **one** benefit of using an embedded system in these machines.

(2)

(ii) Customers use a touch screen to select their destination. They can pay by cash or bank card. Their tickets and a receipt are printed.

The touch screen is controlled by an embedded system.

Give **two** other hardware components in the ticket machine that are controlled by embedded systems.

(2)

1 .....

2 .....

(iii) The ticket machine uses data encryption when a customer pays using a bank card.

State why data encryption is used in this case.

(1)

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P 6 1 8 8 0 A 0 2 0 2 4

2. Question 6 · part c · 6 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

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(b) Compare **four** features of high-level and low-level programming languages.

(4)

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2. Question 6 · part c · 6 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

(c) Artificial intelligence (AI), in many forms, has an increasing impact on our lives.

Discuss this statement considering characteristics, uses and ethical issues.

(6)

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(Total for Question 6 = 15 marks)

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2. Question 6 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part a.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a.i only.

6 (a) Two emerging technologies are DNA computing and quantum computing.

(i) Explain **one** difference between DNA computing and traditional computing.

(2)

(ii) Give a definition of the term **superposition** in quantum computing.

(1)

(b) Ethically, everyone should be able to access the benefits of technology.

Give **two** reasons why this may not be possible.

(2)

1

2

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OLevels.COM SCHOOL

16

P 6 6 4 8 9 A 0 1 6 2 0

3. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 6 · part a.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a.ii only.

6 (a) Two emerging technologies are DNA computing and quantum computing.

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(2)

(ii) Give a definition of the term **superposition** in quantum computing.

(1)

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Give **two** reasons why this may not be possible.

(2)

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4. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 5 · part a.i · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a.i only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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5. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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6. Question 4 · part d.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d.i only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

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6. Question 4 · part d.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d.i only.

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(d) Marie is a researcher in nanotechnology.

(i) State what is meant by the term **nanotechnology**.

(1)

(ii) Give **one** application of nanotechnology.

(1)

(e) Marie has a mobile phone.

The address of the device is:

2024:0eb8:3c4d:0015:0000:0000:1a2f:1a2b

Describe the relationship between the structure of this address and the total number of bits required to store it.

(3)

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6. Question 4 · part d.i · 1 mark · Working and answer

Working and answer

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7. Question 4 · part d.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d.ii only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

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7. Question 4 · part d.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d.ii only.

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(d) Marie is a researcher in nanotechnology.

(i) State what is meant by the term **nanotechnology**.

(1)

(ii) Give **one** application of nanotechnology.

(1)

(e) Marie has a mobile phone.

The address of the device is:

2024:0eb8:3c4d:0015:0000:0000:1a2f:1a2b

Describe the relationship between the structure of this address and the total number of bits required to store it.

(3)

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7. Question 4 · part d.ii · 1 mark · Working and answer

Working and answer

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